

Falling Rates and Rising Superstars

Referee Report

October 30, 2025

Summary

The authors analyze the high-frequency response of firm-level equity prices to FOMC shocks. Using the period from 1994 to 2024, the authors claim that surprise cuts of federal funds target rates have a particularly strong positive impact on industry leaders during periods of low interest rates. They interpret this as evidence that falling rates in low-interest-rate environments favor industry leaders. They also show that industry leaders borrow more, invest more, and acquire more assets than non-leaders in the aftermath of FOMC shocks in low-interest-rate environments. Since the advantage of industry leaders diminishes in high-interest-rate environments, the authors conclude that there exists a “competition-neutral” interest rate at which the effects on leaders and non-leaders are symmetric.

Contribution

The paper contributes to the debate on the effects of interest rates on market concentration. Closely related is the paper by Morlacco and Zeke (2021), who demonstrate that large firms benefit more than small firms in response to an interest rate decline, and that large firms respond to an interest rate decline by increasing their spending on customer capital. The current paper demonstrates that large companies primarily benefit from interest rate cuts in low-interest-rate environments, and they respond by increasing their investment. The authors suggest that their findings align with the theoretical work of Liu, Mian, and Sufi (2022), who demonstrate that a decline in interest rates triggers disproportionate investment responses from large companies, potentially leading to increased market concentration and a decline in aggregate productivity.

Comments

The first part of the referee report considers the robustness of the baseline result. The second part provides general comments.

1. Robustness of the baseline result

I thank the authors for sharing the code and the data. The instructions are very detailed. The summary statistics and the results in the first baseline regression (Table 1, column 1) are aligned with the findings

in other studies. I thus skipped the part on data construction and started my analysis with the generated data.¹

I focus on the baseline results presented in Tables 1 and 2, which I consider to be the core of the paper. The subsequent analysis on the mechanism is also important and perhaps even more interesting, but less vital for the existence of the paper.

The baseline result is that industry leaders benefit from falling rates in low-interest-rate environments more than non-leaders. This result relies on a triple interaction involving variables that capture monetary policy surprises, different interest rate regimes, and whether a firm is an industry leader.

I first confirmed that the provided code exactly reproduces the results reported in the paper. I also checked what is the effect of clustered errors and fixed effects. I found that clustering by FOMC date is important, and it also makes sense, so I retained that approach. Firm fixed effects, however, have only a marginal impact, and I thus eliminated them from my subsequent analysis.

Next, I reviewed the definitions of the main variables and redid the analysis using both the original and modified versions of the variables.

The monetary policy shock *MPS* variable (denoted as *w* in the paper) is calculated in a standard way, and I did not modify this variable.

The dummy variable capturing a low-interest-rate regime, called *POST*, appears a bit ad hoc. The period spans from 1994 to 2024, and the authors define *POST* as zero before 2006 and as one after 2007.² Clearly, interest rates increased substantially in recent years (see figure below). Moreover, Figure 3 in the paper illustrates that the recent period has the largest impact on firm value. This is consistent with some of the most recent evidence, which suggests that monetary policy shocks have had a particularly strong impact post-Covid. To investigate the robustness of results, I define a dummy variable, *LOW* in three different ways:

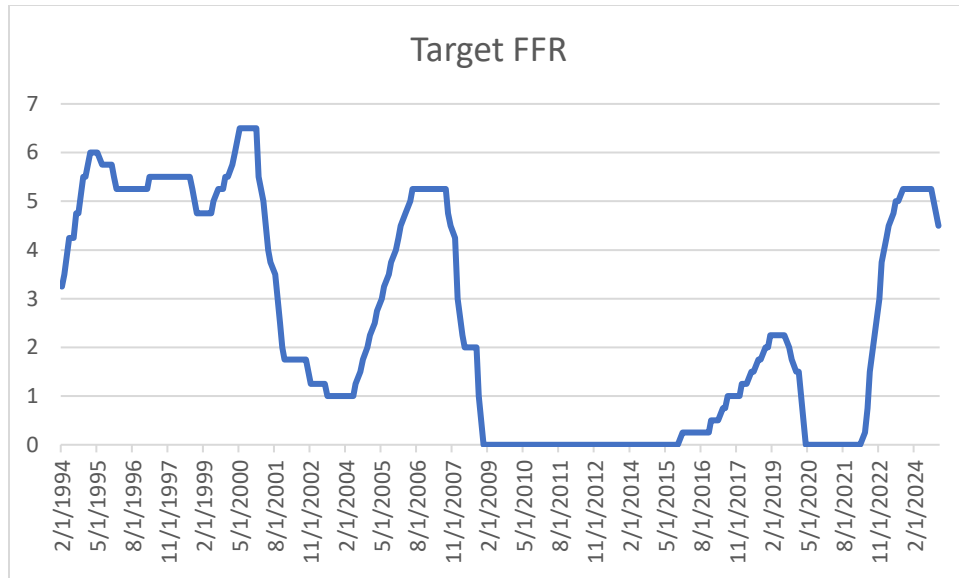
LOW_1 is one if the target rate is below 2.5% (mean value in the data); else, it is zero

LOW_2 is one if the target rate is below 1%; else, it is zero

LOW_3 is one if the target rate is below 0.25% (approximately p75 in the data); else, it is zero

¹ The data-generating step is also very time-consuming; according to the authors, it can take more than two weeks to run the code.

² Splitting the *POST* period into ZLB and non-ZLB diverts attention from the main argument that the effect should depend on the level of interest rate. Reporting the results with ZLB and non-ZLB defined with respect to the whole period (excluding the *POST*) would be more intuitive.



The authors define the variable related to industry leaders either as a continuous variable, X_{bar} (denoting firm rank by average market value percentile across scheduled dates), or a dummy variable, *LEADER* (a dummy variable for a firm being in the top 5% by average market value percentile across all scheduled FOMC dates). To investigate robustness, I redefine *LEADER* in three different ways:

LEADER_1 is one if a firm is in the top 10%; else, it is zero

LEADER_2 is one if a firm is in the top 5% (same as in the authors' analysis); else, it is zero

LEADER_3 is one if a firm is in the top 2.5%; else, it is zero

The analysis in the paper evolves in three steps. First, the authors demonstrate that MPS has a negative impact on firm value, and this effect is stronger in environments with low interest rates. I assess the robustness of these findings in Table 1. A constant term is not tabulated; *t*-statistics with standard errors clustered by FOMC dates are reported in parentheses.

Table 1:

Do monetary policy shocks have a stronger impact on firm value in low-interest-rate environments?

	<i>LOW_1</i>		<i>LOW_2</i>	<i>LOW_3</i>		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>MPS</i>	-0.619	-0.328	-0.685	-0.614	-0.588	-0.592
t-stat.	(-6.80)	(-4.39)	(-4.70)	(-5.46)	(-5.74)	(-4.66)
<i>POST</i>		0.123				
t-stat.		(2.67)				
<i>MPS*POST</i>		-0.687				
t-stat.		(-3.74)				
<i>LOW_x</i>			0.061	0.110	0.125	
t-stat.			(1.23)	(2.08)	(2.07)	
<i>MPS*LOW_x</i>			0.162	-0.025	-0.147	
t-stat.			(0.91)	(-0.14)	(-0.63)	
<i>FFR</i>						-0.019
t-stat.						(-1.79)
<i>MPS*FFR</i>						-0.006
t-stat.						(-0.15)
N	795,059	795,059	795,059	795,059	795,059	795,059
R-square	0.01	0.01	0.01	0.01	0.01	0.01

The first two columns replicate the findings from the paper. The only difference is that I do not use firm fixed effects. Results confirm that *MPS* has a negative impact on firm value, and this effect is strongest during the post-period.

In columns (3) to (5), I redo the analysis using my definitions of interest-rate regimes. I no longer find that the impact of *MPS* is stronger in low-interest-rate regimes. The interaction term between *MPS*LOW_x* is either positive or negative, depending on the exact specification of the variable, and is never significant. I also tried an interaction with a continuous variable, *MPS*FFR*, in the last column and found no significant difference. To assess where the difference in results is coming from, I redefined the authors' *POST* variable by excluding the recent years with elevated interest rates. When I define *POST* as one from 2007 to 2021, and zero elsewhere, the interaction between *POST* and *MPS* becomes insignificant. I conclude that the first claim in the paper is not robust when a low-interest-rate regime is defined based on the nominal target FFR.

The authors discuss in footnote 5 that inflation has been high in recent years, and the real rate remains low. If that is the claim, the authors should then redo the whole analysis using real rates. But then high-frequency shocks should also be defined in relation to real rates rather than nominal rates.

Next, I ask whether industry leaders are more affected by monetary policy shocks than non-leaders. In column 2 of Table 2, I use the authors' continuous variable X_bar to capture firm size. I replicate the author's finding that larger companies have a stronger effect on large companies. The small difference in the results compared to the paper is because I do not use firm fixed effects.

Table 2:

Do monetary policy shocks have a stronger impact on industry leaders?

	<i>LEADER_1</i>		<i>LEADER_2</i>	<i>LEADER_3</i>	
	(1)	(2)	(3)	(4)	(5)
<i>MPS</i>	-0.619	-0.345	-0.604	-0.609	-0.614
t-stat.	(-6.80)	(-3.47)	(-6.43)	(-6.56)	(-6.67)
<i>X_bar</i>		0.018			
t-stat.		(0.66)			
<i>MPS*X_bar</i>		-0.453			
t-stat.		(-3.57)			
<i>LEADER_x</i>			0.008	0.010	0.018
t-stat.			(0.63)	(0.74)	(1.10)
<i>MPS*LEADER_x</i>			-0.088	-0.105	-0.093
t-stat.			(-1.42)	(-1.62)	(-1.38)
N	795,059	795,059	795,059	795,059	795,059
R-square	0.01	0.01	0.01	0.01	0.01

In columns (3) to (5), I redo the analysis using the dummy variable *LEADER*. Regardless of how I define *LEADER_x* (as the top 10%, top 5%, or top 2.5%), I cannot confirm that the effect is statistically stronger for large companies. This is interesting because *LEADER_2* follows the same definition as the variable *LEADER* used by the authors in Table 2.³ I conclude that whether leaders are more exposed to *MPS* depends on whether we use a continuous or dummy variable for industry leaders.

Finally, I assess whether leaders are more affected by monetary policy shocks than non-leaders in low-interest-rate regimes. In the untabulated results, I replicate and confirm the author's findings using a continuous variable for size X_bar and a dummy variable *POST* for a low-interest-rate environment. For robustness, I focus on my definitions of variables *LOW* and *LEADER*. I have three definitions for each variable, resulting in nine possible combinations.

Results are reported in Table 3. As before, the constant is not reported, and *t*-statistics with errors clustered by FOMC date are reported in parentheses. I include all possible interactions and focus my attention on the triple interaction *MPS*LEADER_x*LOW_y*. This interaction term is always negative. It is

³ The authors report the results involving variable *LEADER* in conjunction with the average *FFR* (last column in Table 3).

statistically significant three times (in combination with LOW_3; when the target FFR is below 0.25%) and statistically insignificant six times.

Table 3:

Do monetary policy shocks have a stronger impact on industry leaders in low interest rate environments?

[illegible]

Overall, my analysis does not suggest that monetary policy shocks have a stronger impact in a low-interest-rate regime than in a high-interest-rate environment. It is also unclear whether leaders are more impacted by the monetary policy shocks than followers unconditionally. The analysis does suggest that there may be something special about the effect of monetary policy shocks on leaders in low-interest-rate environments. The magnitude and the statistical significance of these effects depend on how the main variables are defined.

General Comments

The authors motivate their empirical analysis using the Gordon growth model. In the GGM, firm value is:

$$V = \frac{E}{r - g}$$

The authors take the derivative:

$$\frac{d \ln V}{dr} = - \frac{(1 - \frac{dg}{dr})}{(r - g)}$$

They call the expression “*interest rate sensitivity*” and argue that it provides three takeaways:

- (i) The response of firm value to a change in interest rate depends on the level of interest rate r .
- (ii) Firms with stronger growth prospects g respond more strongly to changes in interest rates.
- (iii) The response of firm value to interest rate change depends on possible feedback effects on cashflow growth (dg/dr)

My comments:

- a) The r in the GGM is a firm-specific discount rate, which we typically define as risk premia plus the risk-free interest rate. Is the underlying assumption that the risk premia are constant and do not change in response to FOMC shocks?
- b) The framework predicts that firm growth prospects matter for the change in the firm value. However, g is not used as a control variable in the empirical analysis. Moreover, the “*interest rate sensitivity*” as defined above is closely related to the concept of equity duration. A recent paper by Offner (2025) demonstrates that duration is an important variable in determining a firm's response to monetary policy shocks. It thus seems important to control for growth rate and/or duration.⁴
- c) The most interesting part of the equation is the feedback effect captured by (dg/dr) . For industry leaders, such a feedback effect could be motivated by the theoretical work of Liu et al. However, if the feedback effect is motivated by the theoretical work

⁴ The growth rate can be approximated by the long-term growth rate from IBES. A simple measure of duration could be a payout ratio (readily available measures of duration discussed in Offner (2025) are not available throughout the whole 1994-2024 period).

of Liu et al., why not frame the whole paper as a test of the theoretical work by Liu et al? This would enable sharp predictions, eliminate data snooping concerns, and also allow the authors to differentiate better from the work of Morlacco and Zeke (2021).

References:

Liu, Ernest, Atif Mian, and Amir Sufi, “Low Interest Rates, Market Power, and Productivity Growth,” *Econometrica*, 2022, 90 (1), 193–221.

Morlacco, Monica, and David Zeke, “Monetary Policy, Customer Capital, and Market Power,” *Journal of Monetary Economics*, 2021, 121, 116–134.

Offner, Eric, “Growth vs Value: The Role of Cash Flow Duration in Monetary Policy Transmission,” Working Paper. Universität Hamburg and European Central Bank.